

Ryan Cho

Senior Full Stack Software Engineer

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PROFESSIONAL SUMMARY

Full stack engineer with 8+ years delivering production web applications across finance and investment platforms. Expert in TypeScript, React, and Node.js with PostgreSQL, containerized deployment, and test-driven development. Led REST API design, secure authentication, and scalable SaaS capabilities while partnering with product leaders on architecture and technical strategy in fast-moving remote teams.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Architected and shipped **full-stack TypeScript features** across React and Next.js frontends and Node.js services, supporting portfolio analytics workflows used by **120+ investment professionals** daily.
- Built **RESTful APIs** with structured validation, consistent error handling, and **JWT and OAuth authentication flows**, enabling secure access for internal dashboards and partner integrations.
- Owned **PostgreSQL schema design** and query performance tuning for multi-tenant reporting modules, reducing p95 read latency by **40%** through indexing and connection pooling.
- Established **test-first delivery** with unit, integration, and end-to-end coverage wired into CI pipelines, preventing regressions across **15+ deployable services** before production release.
- Containerized application tiers with **Docker** and automated cloud deployments through **CI/CD pipelines**, enabling repeatable zero-downtime rollouts across staging and production.
- Introduced **Redis caching** for session and reference data, lowering repeated database load by **35%** during peak market activity windows.
- Partnered with product and leadership on **system architecture** for maintainable SaaS modules, influencing roadmap tradeoffs and long-term platform scalability in a remote-first team.
- Configured **Nginx reverse proxies** with SSL/TLS termination for staging environments, improving secure routing between frontend and API layers.

Technology Analyst, Incoming Technology Analyst, Technology Summer Analyst,

May/2019 - Apr/2022

BNP Paribas

Remote | Seventh Avenue, New York

- Developed **TypeScript and Node.js services** for trading operations dashboards, integrating REST APIs with downstream risk systems across **3 business units**.
- Implemented **secure authentication flows** including JWT session management and enterprise SSO, meeting compliance requirements for regulated internal tooling.
- Designed **PostgreSQL data models** and reporting pipelines processing **millions of transaction records** monthly for regulatory and operations teams.
- Maintained **unit and integration test suites** that increased release confidence for production deployments in a regulated banking environment.
- Supported **Docker-based** staging environments and contributed to **CI/CD pipeline** improvements that shortened deployment cycles from days to hours.
- Built **React frontends** with reusable component patterns for analyst workflows, improving task completion efficiency across operations teams.
- Collaborated with architects on **API versioning and validation standards** adopted across technology teams for consistent error handling and contract clarity.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Contributed to **React and TypeScript product features** for a food-tech platform, delivering UI components integrated with REST backend endpoints.
- Wrote **unit and integration tests** alongside feature development, helping the team sustain quality during rapid iteration cycles.
- Assisted with **PostgreSQL query tuning** for order history views, improving responsiveness for high-traffic customer-facing screens.
- Participated in **Docker Compose** local development setups, aligning frontend and API services for consistent engineer onboarding.
- Supported **CI pipeline configuration** for automated builds and test execution on pull requests.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Built **responsive web interfaces** with modern JavaScript for nonprofit program pages, improving accessibility and mobile engagement.
- Integrated **REST APIs** for content and donation workflows with input validation and structured error responses.
- Deployed updates through **Git-based CI workflows**, enabling reliable releases for stakeholder review cycles.
- Collaborated with designers to implement **component-based UI patterns** reusable across campaign landing pages.
- Documented frontend conventions and **testing checklists** that reduced regression issues during iterative content updates.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: TypeScript, React, Next.js, Tailwind CSS, HTML5, CSS3, Component Libraries, Responsive Design

Backend: Node.js, Fastify, RESTful APIs, JWT, OAuth, Express, API Validation

Databases: PostgreSQL, Redis, Schema Design, Query Optimization, Drizzle, Kysely

Cloud & DevOps: Docker, CI/CD, Nginx, SSL/TLS, Cloud Deployment, Turborepo, Monorepo Tooling

Monitoring & Tools: Git, GitHub Actions, Jest, Playwright, ESLint, Prettier, Structured Logging

Workflow & PM: Agile, Code Review, Technical Documentation, Cross-functional Collaboration, Architecture Planning